MIL-T-52463A(ME)
7 September 1971
SUPERSEDING
MIL-T-52463(MO)
23 September 1965

MILITARY SPECIFICATION

TOOL, ALINING, DECK

1. SCOPE

1.1 This specification covers a deck-alining tool used in the erection of the class 60 floating bridge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

Military

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

FSC 5420

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DRAWINGS

ME

C7939 - Tool, Alining, Deck.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A120 - Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses.
A576 - Special Quality Hot Rolled Carbon Steel Bars.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN WELDING SOCIETY

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.18 - Mild Steel Electrodes for Gas Metal-Arc Welding.
B3.0 - Standard Qualification Procedure.
D2.0 - Welded Highway and Railway Bridges.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinoir 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The deck-alining tool shall be in accordance with Drawing C7939 and as specified herein.

3.1.1 Drawing. The drawing forming a part of this specification is an engineering design drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one deck-alining tool for examination within the time frame specified, to prove prior to starting production that his production methods will produce deck-alining tools that comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawing. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Steel bars. Steel bars shall conform to ASTM A576, grade 1040, and shall have a minimum Rockwell hardness of B90.

3.3.2 Steel pipe. Steel pipe shall conform to ASTM A120, hot-dipped galvanized, seamless, standard weight, except that the hydrostatic test is not required and the ends shall not be threaded.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification and size as required.

3.5 Identification marking. The deck-alining tool shall be metal-die stamped as shown on the drawing. Die stamping shall be legible after painting.

3.6 Cleaning, treatment, and painting. The deck-alining tool shall be cleaned, treated, and painted in accordance with MIL-T-704, type B except that the color number designation shall be 34087 in accordance with FED. STD. No. 595.

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3.7 Workmanship.

3.7.1 Steel fabrication. Steel used in the fabrication of the deck-alining tool shall be free from laminations, kinks, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs and rough edges shall be ground smooth.

3.7.2 Welders and welding.

3.7.2.1 Welders. Before assigning any welder or automatic welder equipment operator to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder or operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welding Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX).

3.7.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication "D2.0 - Welded Highway and Railway Bridges". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government

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reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction deck-alining tool shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The deck-alining tools shall be examined as specified herein, for the following defects:

Major

101. Nonconformance to the dimensions shown on the drawing.
102. Material not as specified.
103. Workmanship not as specified.
104. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.

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Minor

201. Identification marking incorrect, missing, or illegible.
202. Cleaning, treatment, and painting not as specified.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105. Containers not as specified for level A or B.
106. Chains not wrapped around the handles and secured with soft annealed wire for level A or B.
107. Strapping not zinc coated for level A.
108. Marking illegible, incomplete, or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified.

5.1.1 Level A. The chains shall be wrapped around the handles and secured with soft, annealed wire. The tools shall be packed together in quantities not to exceed 200 pounds gross weight in close-fitting boxes conforming to PPP-B-601, overseas type, style optional; or PPP-B-621, class 2, style optional. The tools shall be blocked and braced to prevent movement within the boxes. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

5.1.2 Level B. The tools shall be packed as specified in 5.1.1 for level A, except boxes shall be domestic type or class as applicable and strapping need not be zinc coated.

5.1.3 Level C. The tools shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

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5.2 Marking. Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The deck-alining tool is intended for use in pulling the deck-tread panels together during the erection of the bridge to facilitate the installation of the connector pins.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) Level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production deck-alining tools from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish deck-alining tools conforming to this specification.

Custodian:
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Preparing activity:
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Project No. 5420-A222

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-T-52463A(ME) Tool, Alining, Deck

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
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FORT BELVOIR, VIRGINIA 22060

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